

Finger MRI Dictation Cheat Sheet - Expanded Findings & Impressions

Copy-ready language. Replace bracketed text. Use as a workstation reference, not a substitute for surgical correlation or local report style.

Search Pattern

- Name the digit, joint, side, and zone.
- Pulley injury: report A2/A3/A4 and amount of bowstringing.
- Thumb MCP UCL: explicitly state Stener lesion present/absent.

How to Find It — Landmark-Based Identification

Finger/hand MRI is a small-FOV, high-resolution exam where the **axial plane** is your anchor for the pulley system and **sagittal** is your anchor for the extensor mechanism and tendon continuity. Work plane-by-plane, always orienting yourself off the underlying **phalanx** (proximal/middle/distal) and the named joints (MCP, PIP, DIP).

Technique and Planes



PEARL Small-FOV dedicated technique

Image the **single symptomatic ray** with a small FOV (commonly **6–10 cm**), thin slices (**2–3 mm**), and a high-resolution surface/microscopy coil. Always include **T1, fluid-sensitive fat-suppressed** (PD/T2 FS or STIR), and obtain true planes prescribed to the **long axis of the finger**, not the hand.

- Off-axis obliquity is the #1 reason a pulley or collateral ligament looks "torn" — re-check your prescription before calling pathology



HOW TO FIND IT The three planes and what each is for

Use each plane for what it does best:

- **Axial** → pulley system, flexor/extensor tendon cross-section, sagittal bands, neurovascular bundles
- **Sagittal** → extensor mechanism (central slip, terminal tendon), volar plate, flexor tendon continuity, **bowstringing**
- **Coronal** → collateral ligaments, the thumb UCL/RCL, articular surfaces

Flexor Tendons and the Flexor Sheath



HOW TO FIND IT FDP and FDS

On **axial** at the proximal phalanx, the **FDS** lies **superficial (volar)** to the **FDP**. Moving distally, watch the **FDS split** around the FDP — this is the **Camper chiasm** — so the FDS straddles and then passes deep to the FDP to insert on the **volar base of the middle phalanx**.

- **FDP** continues distally to insert on the **volar base of the distal phalanx**
- On sagittal, trace each tendon as a continuous low-signal band from palm to insertion



LANDMARK Camper chiasm

The point where **FDS divides into two slips** that decussate around the FDP, over the **proximal phalanx**. Recognizing it prevents you from mistaking the normal split/partial-volume of FDS slips for a longitudinal split tear.



PITFALL Magic-angle in flexor tendons

Where the flexor tendons curve **~55°** to B0 (e.g., across the MCP/PIP), intratendinous signal can rise on **short-TE** sequences (T1/PD), mimicking tendinosis or a split. Confirm on a **long-TE (T2)** image — true pathology persists; magic angle fades.



HOW TO FIND IT Flexor tendon sheath

Normal sheath is imperceptible; a thin rim of fluid can be physiologic. On **axial fluid-sensitive** images, **circumferential fluid > a thin rim**, synovial thickening, or enhancement suggests **tenosynovitis** (and in the right setting, raises infectious flexor tenosynovitis).

The Flexor Pulley System (Axial is King)



HOW TO FIND IT How to read the pulley system

Scroll **axial** fluid-sensitive images and identify each pulley as a thin low-signal band hugging the flexor tendons against bone. Localize by the **underlying phalanx/joint**, then confirm tendon position relative to bone — pulley failure shows up as **separation of tendon from bone (bowstringing)**.



LANDMARK Which pulley overlies what

Anchor each annular pulley to its bony level:

- **A1** → **MCP joint / metacarpal head–proximal phalanx base** (the trigger-finger pulley)
- **A2** → **proximal phalanx** (long, critical)

- **A3** → **PIP joint**
- **A4** → **middle phalanx** (critical)
- **A5** → **DIP joint**
- Cruciate pulleys **C1 (between A2–A3)**, **C2 (between A3–A4)**, **C3 (between A4–A5)** are thin and collapse to allow flexion



PEARL The critical pulleys

A2 and A4 are the biomechanically essential pulleys — they keep the flexor tendons apposed to the phalanges. Isolated A2 or A4 disruption (classically the **rock-climber's "closed" grip injury**) is what produces clinically significant bowstringing.



MEASURE Recognizing bowstringing

Bowstringing = abnormal increase in the **tendon–bone (tendon–phalanx) distance** as the pulley no longer tethers the tendon. Best shown on a **flexed sagittal** image (volar displacement of the flexor tendons away from bone). A **tendon–bone distance $\geq \sim 2$ mm** in flexion is the commonly cited threshold for a clinically relevant pulley tear; multi-pulley injury exaggerates it.



PITFALL Don't overcall a single pulley

A faint or partial-volume single annular band on one slice is not a tear. Require **secondary signs** — bowstringing, fluid/edema at the pulley, focal discontinuity over **≥ 2 slices** — before calling rupture. The thin cruciate pulleys are normally hard to see.

Extensor Mechanism



HOW TO FIND IT Central slip

On **sagittal** over the PIP, follow the extensor hood to its insertion on the **dorsal base of the middle phalanx** — that's the **central slip**. Disruption (fluid/discontinuity at the dorsal middle-phalangeal base) is the lesion behind a **Boutonnière deformity** (PIP flexion, DIP hyperextension).

- Correlate with axial to see retraction/edema of the central slip



HOW TO FIND IT Terminal extensor tendon

On **sagittal** over the DIP, trace the conjoined lateral bands to the **terminal tendon** inserting on the **dorsal base of the distal phalanx**. Avulsion/discontinuity here = **mallet finger** (DIP droop); look for a small dorsal **bony avulsion fragment**.



HOW TO FIND IT **Sagittal bands**

On **axial at the MCP**, the **sagittal bands** are the transverse fibers that sling the extensor tendon and center it over the metacarpal head. A torn sagittal band (usually **radial** at the long finger — "**boxer's knuckle**") lets the extensor tendon **sublux into the ulnar gutter**, best seen on **axial in flexion**.

- Confirm extensor tendon **off the dorsal apex** of the metacarpal head



PEARL **Match the deformity to the lesion**

A quick map for the extensor mechanism:

- **Mallet finger** → terminal tendon (distal phalanx)
- **Boutonnière** → central slip (middle phalanx)
- **Extensor subluxation / boxer's knuckle** → sagittal band (MCP)

Volar Plate and Collateral Ligaments



HOW TO FIND IT **Volar plate**

On **sagittal**, the **volar plate** is the fibrocartilaginous band on the palmar aspect of the **PIP (and MCP)** joint, anchoring the proximal phalanx to the base of the middle phalanx. In a **dorsal PIP dislocation / hyperextension** injury, look for an avulsion at its distal attachment (± small volar-base fragment).



HOW TO FIND IT **Collateral ligaments**

On **coronal**, the **radial and ulnar collateral ligaments** are paired low-signal bands spanning each joint from the phalangeal/metacarpal head to the adjacent base. Assess continuity and for edema; tears show discontinuity, waviness, and surrounding fluid.



PITFALL **Volar plate calcification mimic**

Post-traumatic **volar plate ossification/calcification** at the PIP can appear as a focal signal abnormality and be misread as an acute fragment — correlate with radiographs and look for cortical/marrow signal.

Thumb Ulnar Collateral Ligament and the Stener Lesion



HOW TO FIND IT **Thumb UCL**

On **coronal** of the thumb MCP, find the **UCL** along the **ulnar** side, running from the metacarpal head to the proximal phalanx base. The normal **adductor pollicis aponeurosis** lies superficial to it. Distal UCL tear with avulsion ("**gamekeeper/skier's thumb**") is the target injury.



LANDMARK The Stener lesion

A **Stener lesion** is a completely torn distal UCL that **retracts and flips superficial/proximal to the adductor aponeurosis**, so the aponeurosis is interposed between the torn ligament and its insertion. On coronal it looks like a rounded **"yo-yo on a string"** mass at the proximal-ulnar MCP.

- This matters because a Stener lesion **cannot heal with immobilization** → it's a surgical lesion



HOW TO FIND IT Thumb RCL

On **coronal**, the **radial collateral ligament** is on the **radial** side of the thumb MCP. Less commonly injured than the UCL, but assess it for discontinuity and instability the same way.

A1 Pulley and Trigger Finger



HOW TO FIND IT A1 pulley / trigger finger

On **axial at the MCP/metacarpal head**, identify the **A1 pulley**; in trigger finger look for **thickening of the A1 pulley** and adjacent **flexor tendon sheath fluid/synovitis**, ± nodular thickening of the flexor tendon that catches during motion.

Neurovascular Bundles



HOW TO FIND IT Digital neurovascular bundles

On **axial**, the paired **digital arteries and nerves** sit at the **volar-lateral corners** of the finger (one radial, one ulnar), just superficial to the flexor sheath. Use them as orientation landmarks and to localize masses (e.g., a **glomus tumor**, classically subungual, or a **digital mucoid cyst** near the DIP).



PITFALL Don't mistake the bundle for pathology

The normal **flowing digital artery** (flow-void / small enhancing tubular structure) and accompanying nerve are constant volar-lateral findings — don't call them a small mass or ganglion. Track them craniocaudally across slices to confirm their tubular, paired nature.

Infection Search Pattern

- **1. Marrow/bone changes:**
- Marrow replacement (darker than muscle/disc on T1; geographic pattern more predictive than reticular/periarticular).

- Marrow edema (isolated finding raises concern in diabetics/high-risk).
- Bone erosion (loss of dark T1/T2 cortical line).
- **2. Surrounding inflammatory changes:** fascia, musculature, subcutaneous fat edema/enhancement.
- **3. Susceptibility suggesting air:** in collections, soft tissues, along fascial planes, in bone.
- **4. Abnormal enhancement:**
 - Rim-enhancing collection = abscess.
 - Superficial lesion without enhancement = ulcer.
 - Enhancement along fascial planes = infectious/inflammatory fasciitis.
 - Non-enhancement of normal structures = ischemia.
 - Other patterns of enhancement may not be specific.
- **5. Differentiating osteomyelitis from neuropathic joint:**
 - Joint effusion with marrow replacement on both sides of joint.
 - Fluid collections/abscesses (periprosthetic, subperiosteal, intraosseous/Brodie's abscess).
 - Sinus tract connecting abnormal marrow signal to skin surface.
 - Disappearance of subchondral cysts on sequential imaging.
 - Progressive marrow/soft tissue changes greater than expected on sequential imaging.

Core Measurements

Structure	Normal / Threshold	Clinical Significance
Stener lesion	UCL retracted proximal and superficial to adductor aponeurosis	Surgical — ligament cannot heal in anatomic position
Pulley bowstringing	>2 mm from phalanx cortex = abnormal	Measured on axial images; >3 mm strongly suggests complete tear
Flexor zone I	FDP insertion to mid-middle phalanx	Jersey finger zone
Flexor zone II	Mid-middle phalanx to distal palmar crease	"No man's land" — FDP and FDS overlap
Flexor zone III	Distal palmar crease to distal carpal tunnel	Lumbrical origin zone
Flexor zone IV	Carpal tunnel	
Flexor zone V	Proximal to carpal tunnel	

Finger MRI Tells

- **A2 and A4** are the two most critical pulleys for finger function; A2 is at the proximal phalanx, A4 at the middle phalanx.

- **Stener lesion:** UCL retracted superficial to the adductor aponeurosis = surgical. If the ligament lies deep to the aponeurosis, conservative management may succeed.
- **FDP avulsion (Jersey finger):** ring finger most common. Look for retracted tendon stump and marrow edema at the distal phalanx base.
- **Central slip disruption** leads to boutonniere deformity (PIP flexion + DIP hyperextension) if untreated.
- **Sagittal band disruption** leads to extensor tendon subluxation at the MCP (boxer's knuckle).
- **Volar plate avulsion:** PIP most common; look for dorsal subluxation and small volar avulsion fragment.

Modified Outerbridge Chondromalacia Grading - Copy-Ready

Use this for any articular surface. Replace [surface/compartment] with the specific location: patellar median ridge, medial femoral condyle weightbearing surface, anterosuperior acetabulum, radiocapitellar joint, tibiotalar dome, first MTP joint, etc.

Grade	MRI shorthand
0	Normal cartilage.
1	Signal heterogeneity/softening or swelling; surface intact.
2	Superficial fissuring/fraying/partial-thickness defect <50% cartilage depth.
3	Deep fissuring/partial-thickness defect >50% cartilage depth, not exposing bone.
4	Full-thickness cartilage loss with exposed subchondral bone, often edema/cysts.

Grade 0 - normal cartilage

F: Articular cartilage of the [surface/compartment] is preserved in thickness and signal without focal fissuring, delamination, or subchondral marrow abnormality.

I: No chondral defect of the [surface/compartment].

Grade 1 - chondral signal change / softening

F: Mild focal cartilage signal heterogeneity/softening of the [surface/compartment] with preserved cartilage thickness and intact articular surface. No subchondral marrow edema.

I: Grade 1 chondromalacia of the [surface/compartment].

Grade 2 - superficial partial-thickness chondral loss

F: Superficial chondral fissuring/fraying of the [surface/compartment] involving less than 50% of cartilage thickness. No full-thickness defect or subchondral marrow edema.

I: Grade 2 chondromalacia of the [surface/compartment].

Grade 3 - deep partial-thickness chondral loss

F: Deep partial-thickness chondral fissuring/defect of the [surface/compartment] involving greater than 50% of cartilage thickness without exposed subchondral bone. [Mild adjacent subchondral marrow edema.]

I: Grade 3 chondromalacia of the [surface/compartment].

Grade 4 - full-thickness chondral loss

F: Full-thickness cartilage loss/fissuring of the [surface/compartment] measuring [X] x [Y] mm with exposed subchondral bone and [subchondral marrow edema/cystic change].

[Small unstable chondral flap/loose body if present.]

I: Grade 4 chondromalacia/full-thickness chondral defect of the [surface/compartment].

Normal / Negative Templates

Normal finger

F: The flexor and extensor tendons of the [digit] are intact without tendinosis, tear, or tenosynovitis. The A2 and A4 pulleys are intact without bowstringing. The collateral ligaments and volar plates of the MCP, PIP, and DIP joints are intact. The articular cartilage is preserved. No marrow edema, fracture, or soft tissue mass. No joint effusion or synovitis.

I: Normal MRI of the [digit].

Useful Caveats

- **Collateral ligament proper vs accessory:** the proper collateral ligament is the primary restraint to lateral stress and is taut in flexion; the accessory collateral ligament inserts on the volar plate and is taut in extension. Report which component is injured.
- **Flexor tendon zones:** surgical planning and prognosis depend on zone of injury. Zone II ("no man's land") tears carry the highest risk of adhesion and poor outcome.
- **Bipartite sesamoid vs fracture** at the thumb MCP: bipartite has smooth corticated margins; fracture has irregular edges with marrow edema.

Pulley / Tendon / Ligament

A2 pulley tear

F: Disruption/attenuation of the A2 pulley at the proximal phalanx with volar bowstringing of the flexor tendons measuring [X] mm from the proximal phalanx cortex, and surrounding soft tissue edema. The flexor tendons are [intact/show secondary tendinosis].

I: A2 pulley tear with flexor tendon bowstringing.

A4 pulley tear

F: Disruption of the A4 pulley at the middle phalanx with focal flexor tendon bowstringing measuring [X] mm from the middle phalanx cortex and peripulley edema.

I: A4 pulley tear.

Multiple pulley tear (A2 + A3)

F: Combined disruption of the A2 and A3 pulleys with diffuse volar bowstringing of the flexor tendons measuring [X] mm from the phalanx cortex extending from the proximal phalanx to the PIP joint level. Surrounding soft tissue edema and tenosynovial fluid.

I: Combined A2 and A3 pulley tears with flexor tendon bowstringing. Surgical consultation recommended given multi-pulley involvement.

FDP avulsion (Jersey finger)

F: Avulsion of the flexor digitorum profundus tendon from its distal phalanx insertion in the [ring/other] finger with proximal retraction of the tendon stump to the level of the [DIP joint/PIP joint/proximal phalanx/palm]. Marrow edema at the distal phalanx base. [Small volar avulsion fragment at the distal phalanx.]

I: FDP avulsion (Jersey finger) of the [digit] with tendon retraction to the level of the [location]. [Leddy and Packer type I/II/III.]

FDS laceration

F: [Partial/complete] discontinuity of the flexor digitorum superficialis tendon at the level of the [zone] with [X] mm gap/retraction and surrounding tenosynovial fluid. The FDP tendon is [intact/also injured].

I: [Partial/complete] FDS tendon tear at zone [zone] of the [digit].

Flexor tendon tear

F: [Partial/full-thickness] tear of the [FDP/FDS] tendon at zone [I-V] with [X] mm tendon gap/retraction and tenosynovial fluid. [Intratendinous signal change consistent with tendinosis at the tear margin.]

I: [Partial/full-thickness] [FDP/FDS] tendon tear at zone [zone].

Central slip tear

F: Discontinuity/irregularity of the central slip insertion at the dorsal base of the middle phalanx with surrounding edema and [small avulsion fragment].

[Lateral band subluxation to the volar aspect of the PIP joint axis, compatible with early boutonniere mechanism.]

I: Central slip extensor tendon injury. [Clinical correlation for boutonniere deformity.]

Extensor sagittal band tear

F: Disruption of the [radial/ulnar] sagittal band at the [MCP] joint with [ulnar/radial] subluxation of the extensor tendon on axial images. Surrounding soft tissue edema and small joint effusion.

I: [Radial/ulnar] sagittal band tear at the [digit] MCP joint (boxer's knuckle) with extensor tendon subluxation.

Volar plate injury

F: [Partial/full-thickness] tear of the volar plate at the [PIP/MCP] joint with capsular edema and [small volar avulsion fragment]. [Dorsal subluxation of the phalanx.]

I: [Partial/full-thickness] volar plate injury of the [joint].

Thumb MCP UCL tear (Stener lesion present)

F: Complete tear of the thumb MCP ulnar collateral ligament at the distal attachment with proximal retraction of the ligament stump superficial to the adductor aponeurosis, consistent with a Stener lesion. Adjacent soft tissue edema and small joint effusion.

I: Complete thumb MCP UCL tear with Stener lesion. Surgical repair indicated.

Thumb MCP UCL tear (non-Stener)

F: [Partial/complete] tear of the thumb MCP ulnar collateral ligament at the [proximal/distal] attachment. The ligament remains deep to the adductor aponeurosis without Stener lesion. Surrounding edema and small joint effusion.

I: [Partial/complete] thumb MCP UCL tear without Stener lesion.

[May be amenable to conservative management pending clinical correlation.]

Thumb MCP RCL tear

F: [Partial/complete] tear of the thumb MCP radial collateral ligament at the [proximal/distal] attachment with periligamentous edema and small joint effusion. [No significant lateral subluxation.]

I: [Partial/complete] thumb MCP radial collateral ligament tear.

Collateral ligament injury (PIP)

F: [Partial/complete] tear of the [radial/ulnar] collateral ligament of the [digit] PIP joint at the [proximal/distal] attachment with periligamentous edema and small joint effusion. [Mild lateral subluxation on coronal images.]

I: [Partial/complete] [radial/ulnar] collateral ligament tear of the [digit] PIP joint.

Soft Tissue Masses

Giant cell tumor of tendon sheath

F: Well-circumscribed, lobulated soft tissue mass along the [flexor/extensor] tendon sheath of the [digit] measuring [X] x [Y] x [Z] mm, demonstrating low-to-intermediate T1 and low T2 signal with susceptibility artifact from hemosiderin deposition. [Mild enhancement.] No osseous invasion or aggressive features.

I: Soft tissue mass along the [digit] tendon sheath, characteristic of giant cell tumor of tendon sheath (GCTTS/PVNS). Excision may be considered if symptomatic.

Mucous cyst

F: Small, well-defined cystic lesion at the dorsal aspect of the [digit] DIP joint measuring [X] mm, demonstrating T1 low signal and T2 high signal, contiguous with the DIP joint and adjacent to a marginal osteophyte.

[Mild overlying skin thinning.]

I: Mucous cyst (digital myxoid cyst) of the [digit] DIP joint associated with underlying osteoarthritis.

Enchondroma

F: Well-defined intramedullary lesion in the [proximal/middle] phalanx of the [digit] measuring [X] mm, demonstrating lobulated T1 low/intermediate signal and T2 high signal with characteristic chondroid matrix. No cortical destruction, periosteal reaction, or soft tissue extension. [Faint internal mineralization.]

I: Enchondroma of the [digit] [proximal/middle] phalanx. Benign appearance.

[If pain is present, consider pathologic fracture risk.]

Source Notes

Built as a practical dictation aid from standard MSK MRI reporting patterns, review-level radiology references, and the existing shoulder cheat-sheet style. Use surgical correlation and local report conventions when needed.

Test Yourself & Visual Pearls

A rapid self-check on the tendons, pulleys, and ligaments that make or break hand function — read the question, then expand for the answer.

Key points



KEY POINTS Hand essentials

- Flexor pulleys A2 (proximal phalanx) & A4 (middle phalanx) are the critical pulleys; rupture → bowstringing of the flexor tendons away from bone.
- Thumb UCL tear with a Stener lesion is surgical — the torn ligament retracts superficial to the **adductor aponeurosis**, which becomes interposed and prevents healing.
- FDP avulsion from the distal phalanx = **jersey finger**, graded by the **Leddy classification** based on level of tendon retraction.
- Mallet finger = terminal (distal) extensor tendon injury at the DIP; boutonnière = central slip injury at the PIP — distinct levels, distinct deformities.
- Sagittal band rupture (typically radial, long finger) = **boxer's knuckle**, allowing extensor tendon subluxation at the MCP.

FLEXOR PULLEY SYSTEM (lateral finger)



A2 (proximal phalanx) & A4 (middle phalanx) = critical pulleys

rupture → bowstringing = increased tendon–bone distance

Flexor pulley system. Annular pulleys A1–A5 hold the flexor tendons against the phalanges. **A2 (proximal phalanx) and A4 (middle phalanx) are the critical pulleys** — rupture causes bowstringing (increased tendon–bone distance), the climber's injury.

Rapid review

? QUIZ A rock climber has pain and a palpable cord along the volar finger; MRI shows increased tendon–bone distance. Which structures are torn and what is the sign?

? QUIZ FDP tendon avulses from the volar base of the distal phalanx. What is the eponym and how is it classified?

? QUIZ A patient cannot actively extend the DIP and holds it flexed after a "jammed" fingertip. Which tendon is injured?

? QUIZ Disruption of the central slip at the PIP produces what deformity, and what do the lateral bands do?

? QUIZ A boxer has dorsal MCP pain with extensor tendon snapping. What is torn and which direction does the tendon go?

? **QUIZ** Skier falls on an abducted thumb; MRI shows the torn UCL balled up and displaced proximal to the adductor aponeurosis. Name the lesion and why it matters.

! **PITFALL** Don't call every proximal UCL stump a Stener lesion. The diagnosis requires the retracted ligament to lie **superficial to the adductor aponeurosis**; a torn but non-displaced UCL still sitting deep to the aponeurosis is not surgical.

? **QUIZ** A finger forced into hyperextension dislocates dorsally at the PIP. Which volar structure fails?

? **QUIZ** A diabetic patient has painful catching of a finger that locks in flexion. What is the lesion and where?

? **QUIZ** A small, exquisitely painful subungual nodule shows intense post-contrast enhancement on MRI. Diagnosis?

? **QUIZ** A firm volar finger mass along the tendon sheath is low signal on all sequences and "blooms" on gradient-echo. What is it?

! **PITFALL** On gradient-echo, **hemosiderin blooming** is the giveaway for giant cell tumor of the tendon sheath — but correlate with the typical periarticular/tendon-sheath location and lobulated morphology.